Assignment 9 - Doxygen

TW Lab 2025

In today's lab, you are required to create the documentation of a small multi-file C++ project using *Doxygen*. The project files are attached. Some famous projects, such as OpenCV, LLVM and D-Bus use Doxygen for documenting their source code.

1 Files

Following project files have been provided to you. You are required to generate the docs using Doxygen.

- **Logger.h** - Defines a simple Logging class.
- **CalculationEngine.h** - Defines a class for mathematical operations, which uses the `Logger`.
- **main.cpp** - The application entry point.
- **Doxyfile_template** - A sample Doxyfile

2 Instructions

Follow these instructions and complete all the tasks given below:

2.1 Install doxygen

Use *apt* to install doxygen on Ubuntu:
`sudo apt install doxygen`

2.2 Complete the documentation

For each source file, find the *TODO* comments and all undocumented elements. Write correctly formatted Doxygen comments using the special commands (@, \) for the following elements:

- Main page - Use $\backslash mainpage$ or $@file$ to define the project overview.
- Logger class - Use $@file$, $@class$, $@brief$.
- Log_Level enum - Use $@enum$ and document all enumerator values.
- Logger::Logger - Document the constructor and its parameters.
- CalculationEngine class - Use $@file$, $@class$, $@brief$, $@detail$.
- CalculationEngine::power_calculation - Add $@param$ for *base* and *exponent*, and $@return$ for the result. Review the existing $\backslash todo$ tag.

2.3 Generate HTML & PDF documentation

- Create a Doxyfile with appropriate settings that are required to generate HTML and LaTeXoutputs from these source files. You may use the template file given to you as a starting point.
- Generate HTML using the command:
`doxygen Doxyfile`.
- Navigate to the `doc_latex` directory and run the command `make` to generate a PDF.

3 Submission

Upload the following to Google Classroom:

1. The three completed source files
2. The final Doxyfile used for generation
3. A screenshot of generated HTML documentation
4. Generated PDF file